

**Seqela**

BANKING TECHNOLOGY · ETHIOPIA

PREPARED FOR

Development Bank of Ethiopia

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Server specification — on-premises deployment of DBE iBOC

Hardware, storage, network, and environment sizing for the Integrated Back Office Automation and Credit Intelligence Platform. Data stays in DBE's data centre. Core banking stays DBE's.

DOCUMENT

Infrastructure specification

PREPARED BY

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REFERENCE

SEQ/DBE/IBOC/INFRA/2026-09

AUDIENCE

DBE IT, Infrastructure, Security

DATE

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Ask of DBE IT: provision the environments below (UAT now; production before go-live). Seqela supplies the application, license key, and install runbook. DBE supplies servers, storage, TLS, DNS, firewall, SMTP, NTP, backup target, and CBS connectivity when ready. Do **not** size production on the 2 vCPU / 4 GB lab minimum used for a DECSI pilot host.

1. Purpose and deploy model

iBOC is the credit back-office for DBE: origination, KYC/documents, appraisal, GPS collateral, committees, seven financing product engines, and post-disbursement monitoring. It runs **on-premises or in DBE private cloud**. It is not multi-tenant SaaS. There is no vendor phone-home.

UAT / training

Configuration, UAT, and training-of-trainers. One VM is enough. Keep it off the production host.

Production

Live system for HQ and regional offices. Split application and database. Two app nodes preferred.

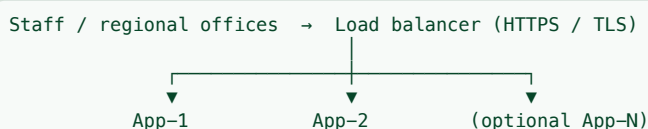
DR (recommended)

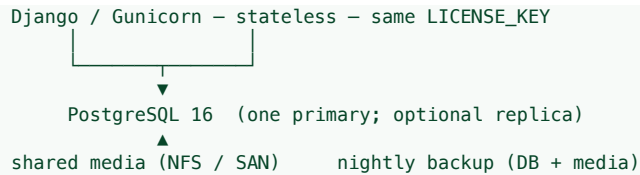
Cold standby or streaming replica. Restore drill at least twice a year.

2. UAT server (single host)

All-in-one Docker Compose: application, PostgreSQL 16, document media, nightly backup.

ITEM	SPECIFICATION
Count / role	1 virtual machine
CPU	8 vCPU
RAM	32 GB
OS disk	100 GB SSD
Data disk	500 GB SSD (database + documents / photos)
Operating system	Ubuntu 22.04 LTS or RHEL 8/9, x86_64
Software on host	Docker Engine 24+, Docker Compose v2, Git, NTP
Network	LAN to HQ / UAT users; static IP or DNS name

3. Production architecture



If two app nodes each keep local disks, a file uploaded on node A is missing on node B. Shared media is mandatory for HA.

4. Production servers

4.1 Application nodes (2 for high availability)

Stateless web tier. OCR (Tesseract English + Amharic), PDF packs, and document upload run here. Do not put PostgreSQL on these boxes.

ITEM	PER NODE
Count	2 behind DBE load balancer (nginx / F5 / HAProxy)
CPU	8–16 vCPU
RAM	32 GB (64 GB if concurrent OCR / PDF is heavy)
OS disk	100 GB SSD
OS / runtime	Ubuntu 22.04 LTS or RHEL 8/9 · Docker 24+ · Compose v2
Gunicorn (start point)	8 workers × 2 threads · timeout 120 seconds

4.2 Database server

ITEM	SPECIFICATION
Count	1 primary (+ streaming replica if DBE requires database HA)
CPU	8–16 vCPU
RAM	32–64 GB
Data disk	500 GB–1 TB SSD, dedicated — not NFS
Engine	PostgreSQL 16
Encryption at rest	Volume / disk encryption per bank / INSA policy

One logical primary only. Do not run two independent databases. A replica is for failover or reporting — not a second write copy.

4.3 Shared document storage

KYC packs, collateral photos, GPS evidence, signatures, and implementation–visit images.

ITEM	SPECIFICATION
Type	NFS or SAN, same path on every app node
Start size	1 TB
Growth	Plan 2–5 TB over 3 years (project dossiers + site photos)
Backup	Included in the nightly job together with the database

4.4 Load balancer / reverse proxy

Use DBE's existing F5, nginx, or HAProxy.

ITEM	SPECIFICATION
TLS	Terminate HTTPS with the bank certificate
Health check	GET /hub/ — node must be able to reach PostgreSQL
Upload body limit	≥ 50 MB (PDFs and field photos)
Sticky sessions	Not required (sessions stored in PostgreSQL)

4.5 Disaster recovery (recommended)

ITEM	SPECIFICATION
Option A	Cold VM matching production database size; restore from backup
Option B	Streaming replica in a second site / data centre

ITEM	SPECIFICATION
RPO	≤ 24 hours with nightly backup; minutes with a replica
RTO	4–8 hours (cold restore); under 1 hour if a replica is kept warm
Restore drill	At least twice a year

5. Bill of materials for DBE IT

5.1 Recommended production (HA)

SERVER	CPU	RAM	DISK
App-1	8–16 vCPU	32 GB	100 GB OS SSD
App-2	8–16 vCPU	32 GB	100 GB OS SSD
Database primary	8–16 vCPU	32–64 GB	500 GB–1 TB data SSD
Database replica (optional)	Same as primary	Same	Same
Shared media	—	—	1 TB+ NFS / SAN
Backup / DR target	—	—	2–4 TB off-box

5.2 Minimum production (no HA)

Acceptable only if HA is deferred. Still split application and database.

SERVER	CPU	RAM	DISK
Application	8 vCPU	32 GB	100 GB OS + 1 TB media
Database	8 vCPU	32 GB	100 GB OS + 500 GB data SSD
Backup target	—	—	2 TB off-box NAS / tape / object

Short HQ pilot only: if a single box is all that can be issued at first, the floor is **16 vCPU · 64 GB · 1 TB SSD**. That is a pilot host, not the national production shape. UAT and production must not share that host.

6. Network, ports, and integrations

DIRECTION	PORT	USE
Staff / regional offices → load balancer	443/tcp	HTTPS staff hub, Digital Apply, field tablets
App → PostgreSQL	5432/tcp	Internal VLAN only — never internet
App → SMTP	25 / 587	Password reset and notifications
App → DBE CBS (later)	Bank-defined	Customer lookup and disbursement booking
App → NTP	UDP 123	Required for MFA (TOTP)
Outbound HTTPS (optional)	443	Maps / geocoding, OS and Docker patches; AI only if policy allows

Firewall: staff and branches reach 443 only. PostgreSQL, Docker, and admin ports stay on the data-centre VLAN, jump host, or VPN. **DNS:** a name such as `iboc.dbe.et` (or DBE naming). Static IPs on all servers.

Workstations: current Chrome, Edge, or Firefox; 1920×1080 preferred for appraisal and collateral. Field GPS / camera requires HTTPS.

7. Software stack on the servers

COMPONENT	ROLE
Django + Gunicorn	Staff hub, product engines, committees, Credit Intelligence
PostgreSQL 16	System of record
Tesseract OCR (eng + amh)	Scanned documents
Media volume	KYC, collateral, implementation photos
Nightly backup	Database dump + media → off-box store (14+ day retention)
License key	Offline SEQLA1. ... key; verified locally; no phone-home

Host does not need a separate Python or PostgreSQL install when using Docker Compose. Production runs with `DEBUG=False` and `USE_GUNICORN=1`.

8. Split of responsibility

DBE PROVIDES	SEQELA PROVIDES
Virtual machines or bare metal; storage; TLS certificate; DNS; firewall; SMTP; NTP; backup target; jump host / VPN; CBS connectivity when ready.	Application images; signed license key; install runbook; database migrations; UAT support; Year-1 hypercare.

Hardware, data-centre, and connectivity cost remain with DBE.

9. Security baseline (hosting)

- TLS on every staff and tablet path; secure cookies; MFA (TOTP) for staff.
- PostgreSQL not published to the internet; unique strong `SECRET_KEY` and database password per environment.
- Disk / volume encryption for database and media; encrypted off-box backups.
- Clock synchronized (NTP) — MFA codes fail if the server clock drifts.
- Django Admin reachable only from jump host / VPN.
- UAT and production use different secrets, license keys, and databases.

10. Pre-provision checklist

	ITEM	OWNER
☐	UAT VM sized per §2	Infrastructure
☐	Production app + database hosts sized per §4 / §5	Infrastructure
☐	Shared media mount (if two app nodes)	Infrastructure
☐	Off-box backup target with restore path	Infrastructure
☐	DNS name + static IPs + TLS certificate	Infrastructure / Network
☐	Firewall: 443 to staff; 5432 internal only	Network / Security
☐	SMTP and NTP available from app hosts	Infrastructure
☐	Jump host / VPN for admin access	Security
☐	CBS endpoints and credentials (when cutover is approved)	CBS / IT
☐	Seqela license key and release package received	Seqela / DBE IT

This note sizes the platform for HQ and regional-office use. Final VM counts can be confirmed with DBE IT when the deployment is scheduled. It is a working specification, not a contract or sign-off pack.

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Companion documents: DBE IBOC proposal (SEQ/DBE/IBOC/2026-09). Core banking remains DBE's. This note covers hosting only — not commercial terms.